

Problem Set - 9

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1 Mechanics

1.1 Friction Transmission

A long cylindrical roller of radius R_0 , rotating around its axis with angular velocity ω_0 , is pressed against a freely rotating (without friction in the axis) disk of radius R . The line of contact between the disk and the roller coincides with the radius of the disk.

1) Find the steady-state angular velocity ω_μ rotation of the disk if the friction between the roller and the disk is dry.

2) Find the steady-state angular velocity ω_η rotation of the disk, if the friction is viscous. Consider that the magnitude of the viscous friction force per unit length of contact is proportional to the relative speed of movement of the contacting surfaces of the roller and disk.

3) Define the ratio $k = \omega_\eta/\omega_\mu$.

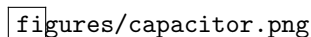
2 Electromagnetism

2.1 Not-so-parallel Plate Capacitor

A capacitor is formed by two rectangular conducting plates having edges L_1 and L_2 . The plates are not parallel. One pair of edges of length L_1 is separated by a distance d_1 everywhere, and the other pair of edges of length L_2 is separated by d_2 everywhere; $d_2 > d_1$. (A Guide to Physics Problems, 3.16)

a) Neglecting edge effects, when a voltage difference V is placed across the two conductors, find the potential everywhere between the plates.

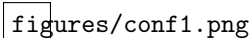
b) Determine the capacitance.

figures/capacitor.png

2.2 Applications of the Method of Conformal Mapping

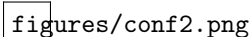
2.2.1 Potential above Half Planes

An infinite conducting plane is divided by the line $z = 0$. For $x > 0$ the potential in the plane is $+V_0$ while for $x < 0$ the potential is $-V_0$. Evaluate the potential everywhere. (A Guide to Physics Problems, 3.25)

figures/conf1.png

2.2.2 Potential of Halved Cylinder

Consider an infinitely long conducting cylinder of radius a with its axis coinciding with the z -axis. One half of the cylinder (cut the long way) ($y > 0$) is kept at a constant potential $+V_0$, while the other half ($y < 0$) is kept at a constant potential $-V_0$. Find the potential for all points inside the cylinder and the field $\vec{E}$ along the z -axis. (A Guide to Physics Problems, 3.26)

figures/conf2.png

3 Optics

3.1 Refractive Pyramid

A regular triangular pyramid $ABCD$ is made of a transparent material with a refractive index of $n = 1.6$. All angles at the top of the pyramid A are right angles. At the base of the pyramid lies a regular triangle BCD . This pyramid can also be represented as a “corner cut off from a cube”, whose side lengths are equal to $a = 2.0$ mm. The pyramid is uniformly illuminated so that the light falls along its height, perpendicular to the plane of the base. (Zhautykov 2013, P1.C)

- 1) Draw the area on the base of the pyramid illuminated by the light refracted by face ABC .
- 2) A screen is located parallel to the base of the pyramid at a distance of $L = 10$ cm from it. Draw the areas illuminated by the light refracted by the pyramid on this screen. Indicate the position and dimensions of these areas.

figures/pyramid.png

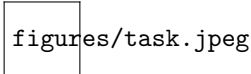
3.2 Cassini Division

In an opaque flat layer of thickness h a hole is drilled, the shape of the side surface of which is a truncated cone with base diameters d and D , the axis of which is perpendicular to the flat surfaces of the layer. The side surface of the cone is silvered and perfectly reflects the light falling on it. Behind the layer at a distance L from the nearest flat surface of the layer there is a flat screen. (RuPhO 2023)

- 1) Draw a picture of what you see on the screen. Indicate all the characteristic geometric dimensions in the drawing.
- 2) How will the picture from point 1 change if a thin collecting lens is placed between the layer and the screen? The focal length of the lens $F = L/2$, the main optical axis of the lens coincides with the axis of the cone, the screen is located in the focal plane of the lens.

3.3 Inhomogeneity in a Prism

Experimenter Gluck conducts optical studies with a solid prism, cross section OAB which is a right triangle with legs AB and $OA = h$. All faces of the prism are either parallel or perpendicular to the plane of the drawing.



If we introduce a rectangular coordinate system (x, y) with origin at point O as shown in the figure, the refractive index of the prism material depends only on the coordinate x as:

$$n(x) = \frac{3}{2 - x/h}$$

Gluck decided to completely illuminate the face of the prism containing OA with a light beam perpendicular to it. The bottom of the prism, containing AB is covered with a substance that completely absorbs the light that hits it. The prism is located in air, the refractive index of which can be considered equal to one. (RuPhO 2024)

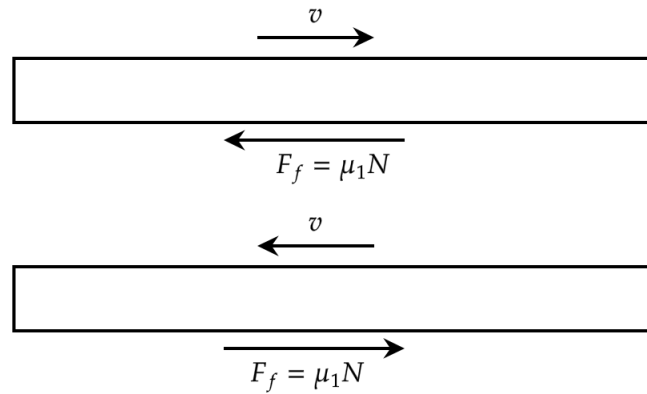
- 1) Let us consider a ray entering a prism at a point with coordinate x_0 . Find the equation of the trajectory of this ray inside the prism before it hits the sides AB and OB .
- 2) At what angle values $\angle AOB = \alpha$ do the rays of light hit all points on the side OB prism sections?
- 3) At what angle values α all rays that reached the side OB sections of a prism are refracted and pass into the air?

4 Miscellaneous

4.1 Creeping Strip

By using a special technology for processing polymeric materials, it is possible to achieve anisotropy of their surface properties. For example, the movement of a polyethylene strip along the horizontal surface of a table in the direction $\overrightarrow{AB}$ (to the right in the figure) is characterized by the coefficient of friction μ_1 , and the movement in the direction $\overrightarrow{BA}$ - coefficient of friction $\mu_2 < \mu_1$. If you subject a strip of length lying on the table to l cyclic heating and cooling, you can see that the strip moves along the surface of the table.

(RuPhO 2021, 10-1)



- 1) In which direction will the strip shift with a large number of cycles?
- 2) How far will the strip move in N heating-cooling cycles if the difference between the maximum and minimum temperatures in the cycle is equal to ΔT ?

Note: The strip has a constant thickness, does not bend and is pressed to the table evenly with its entire surface. Heating and cooling occur slowly.